2020 Condition report

Bridges and Major Culverts

Uganda National Roads Authority

Network Planning

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1 INTRODUCTION

Bridges are an integral part of the road network and therefore require optimum management to preserve the road network. They are a key element with a high asset value and as such require efficient care to minimize losses. The failure of a bridge also poses a high safety risk to road users and highly disrupts traffic. As a result, attention must be given to the proper management of bridges and major culverts on the management.

The first step to a proper management system is regular inspection followed by timely maintenance. This is to ensure the safety of the road users and to protect the capital investment of the structure with a minimal operational cost.

This document reports on the condition of the bridges and major culverts which will inform the maintenance needs of the structures on the network.

2 UNRA STRUCTURES

UNRA classifies its structures as Bridges and Major Culverts.

2.1 Bridges

A structure is classified as a bridge if one or more of the following is true:

- Any single span, measured horizontally at the soffit along the road centreline between the faces of its supports >= 6 m, or
- The individual clear spans > 1.5 m, and overall length between abutment faces > 20m, or
- Opening height >= 6, or
- Total cross-sectional opening >= 36m2, or
- Structure is a road-over-rail (even if span < 6m)
- If the structure consists of clearly identifiable elements such as deck slabs, deck expansion joints, abutments, piers and foundation footings. Elements such as invert slabs, apron slabs, cut-off walls are normally not present. Should a structure with dimensions of a Major Culvert present mostly bridge elements, then it should be considered a bridge and recorded as such in the BMS.

Bridges are named as Bxxx with xxx denoting the unique number assigned to each bridge.

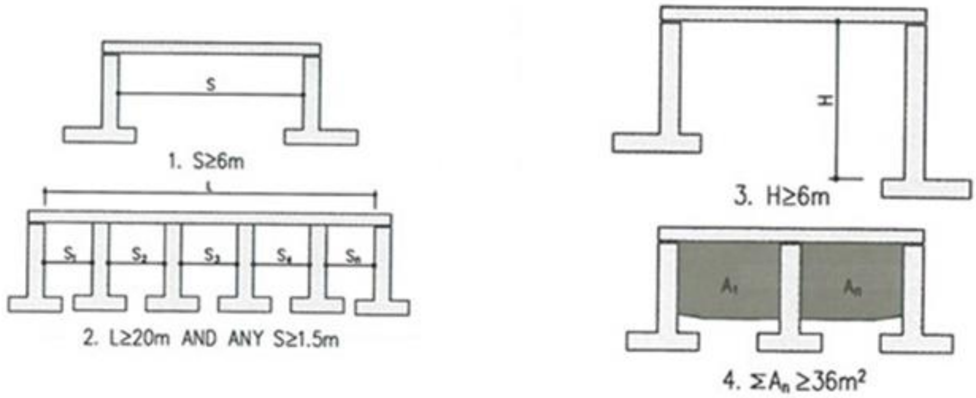


Figure 2-1: Definition of a Bridge

2.2 Major Culverts

A Major Culvert is a structure, normally “buried” with road fill or road slabs on top, consisting of “cellular” units. A cellular unit can typically be described as an “opening” where, in general, the overall cell length is greater than the cell width. Elements such as separate deck slabs, abutments/piers, foundations, etc. are not clearly identifiable while elements such as invert slabs, apron slabs, cut-off walls etc. are normally present. Elements such as separate deck slabs, abutments/piers, foundations, etc are NOT clearly identifiable. Should a structure have dimensions exceeding that of a Major Culvert but presenting mostly culvert elements, then it should be considered a Major Culvert and recorded as such in the BMS.” Culverts with a pipe span of less than 2m are termed Minor Culverts and are not included in the BMS.

Major Culverts are named as Cxxx with xxx denoting the unique number assigned to major culvert.

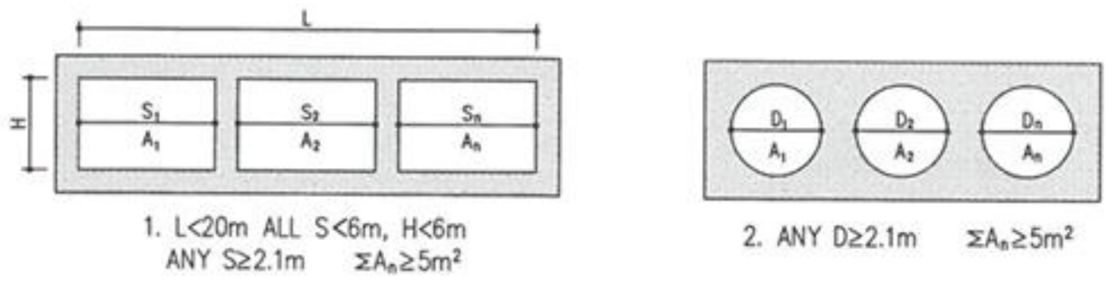


Figure 2-2: Definition of a Major Culvert

According to the UNRA Bridge management System (BMS), the total number of structures on the network is 727 with 434 bridges (60%) and 293 Major Culverts (40%). From Figure 2-4, it is observed that the majority of structures are located on unpaved roads with 460 on unpaved roads and 267 on paved roads.

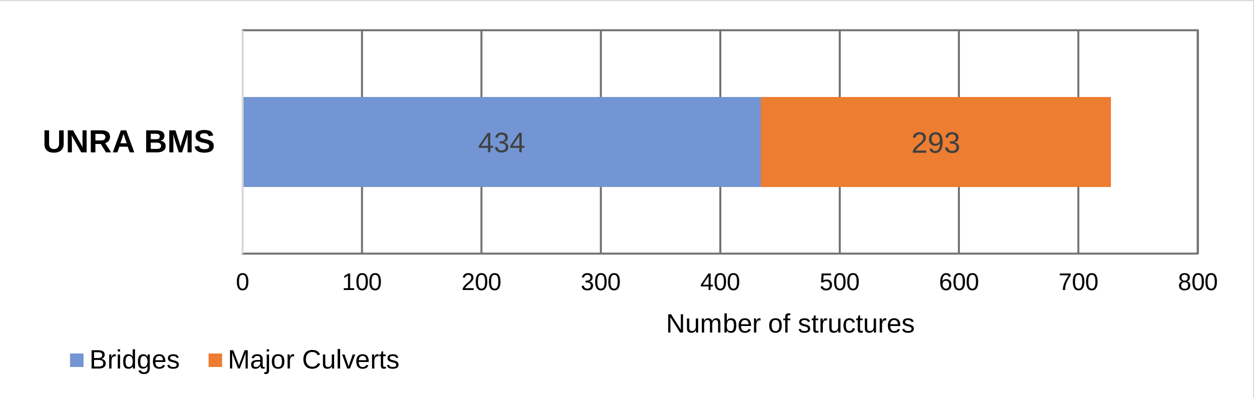


Figure 2-3: Split between Bridges and Major Culverts on the UNRA Network, 2020

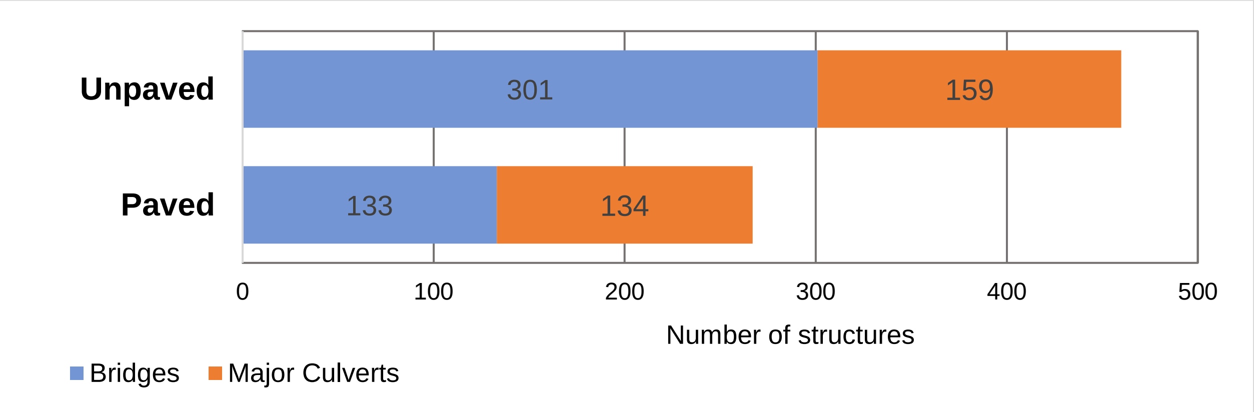


Figure 2-4: Split between Bridges and Major Culverts on the Paved and Unpaved Network, 2020

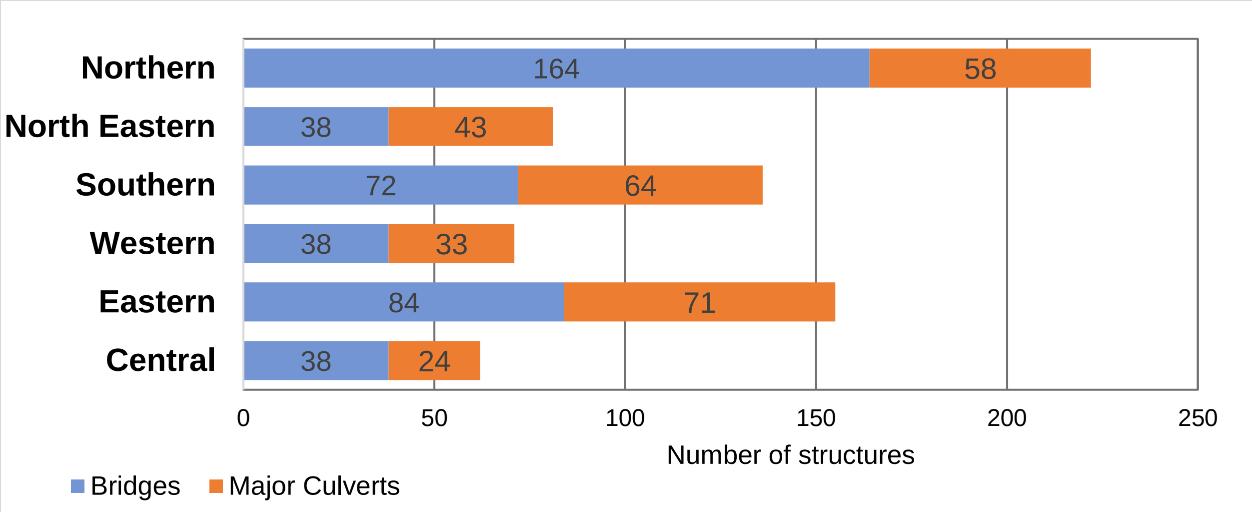


Figure 2-5: Split between Bridges and Major Culverts per region, 2020

From Figure 2-5 above, it can also be observed that the Northern region followed by the Eastern and Southern regions have the highest numbers of structures while the Central region has the smallest number of structures. This has an impact on the funding levels for routine maintenance of structures per region.

3 CONDITION OF BRIDGES AND MAJOR CULVERTS

Visual inspections are carried out to ascertain the inventory of the structure, identify defects, the stage and extent of the defect and the maintenance recommendations. Inspections are carried out by qualified bridge inspectors and must consist of a thorough examination of all the components of the structure. For bridges, the components assessed are; Waterways (if applicable), Foundations, Substructure (Abutments and Piers), Bearings, Superstructure, Non-structural elements (joints, drainage, parapets etc.) and Approaches. For Major Culverts, the components assessed are; Waterways, Inlets and Outlets, Substructure and Superstructure (Cell Invert, Cell Roof Slab, Cell walls), Precast Pipe Culvert, Corrugated Steel Pipe and Roadway.

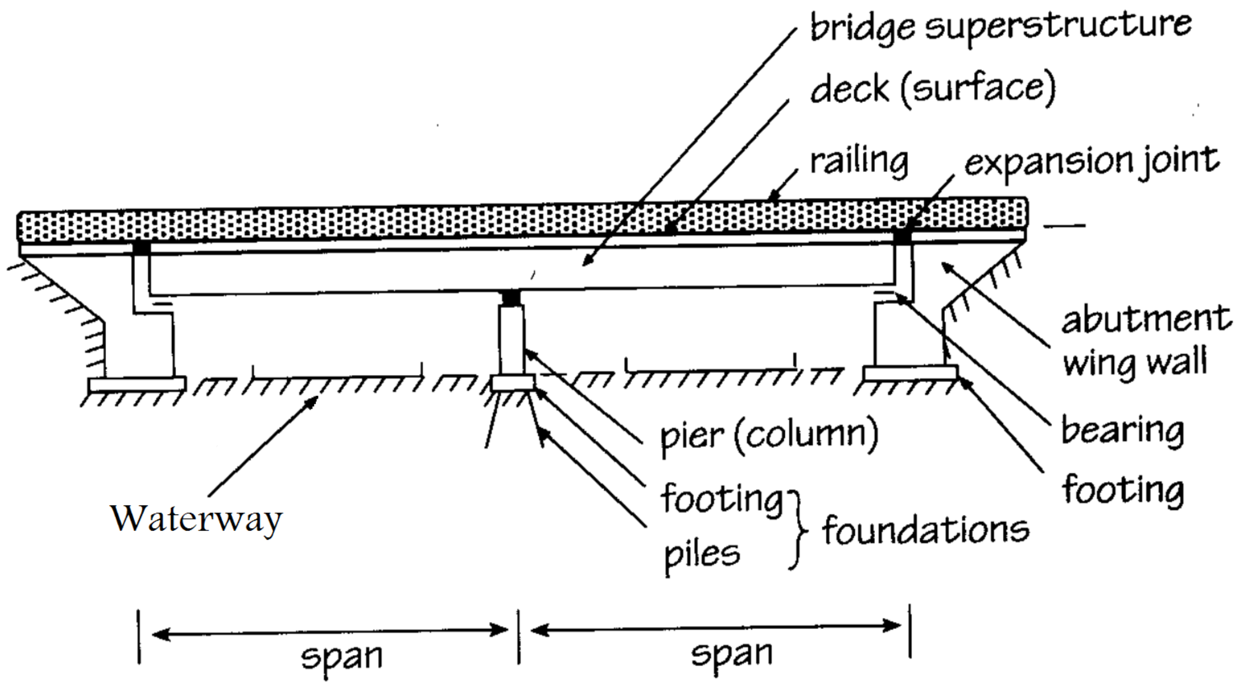


Figure 3-1: Components of a bridge

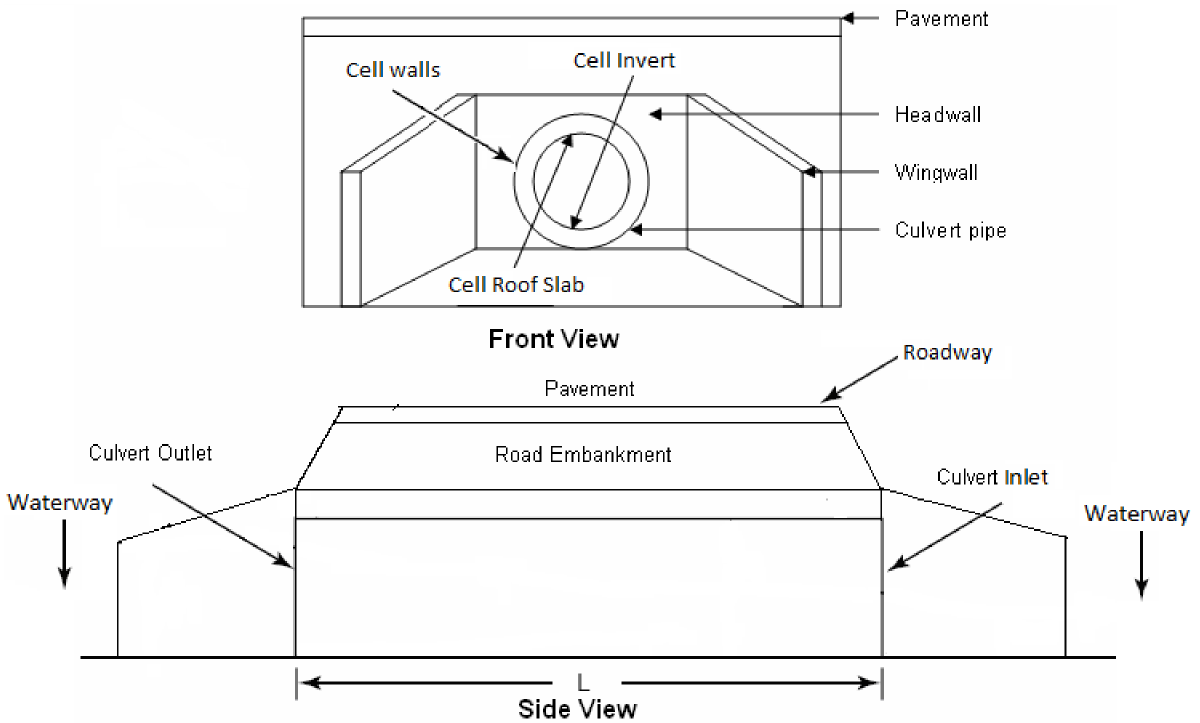


Figure 3-2: Components of Major Culverts

A numerical scale (0-9) is used to rate the condition of both the main components and subcomponents. These ratings are based on the current prevailing condition of the component and are used to calculate an overall rating of the structure. The

Overall Rate calculated is classified in 5 condition categories, i.e. critical, poor, marginal, satisfactory, good, as follows;

Table 3-1: Condition categories of the Overall Rating

Calculated Overall Rate	Condition Category
0 and 1	Critical
2 and 3	Poor
4 and 5	Marginal
6	Satisfactory
7 to 9	Good

3.1 Condition of Bridges

Inventory and condition assessments were first done in 2008 by a consultant. This data populated the first database on which all the subsequent assessments have been based. Other inspections were done in 2011 and 2015 with the most recent one being done in FY 2017/18 to 2018/19 by UNRA staff and Data Collection Consultants. A total of 385 (88%) Bridges on the network have condition assessment data. Based on these visual assessments, the following conclusions can be made with regards to the condition of Bridges in UNRA.

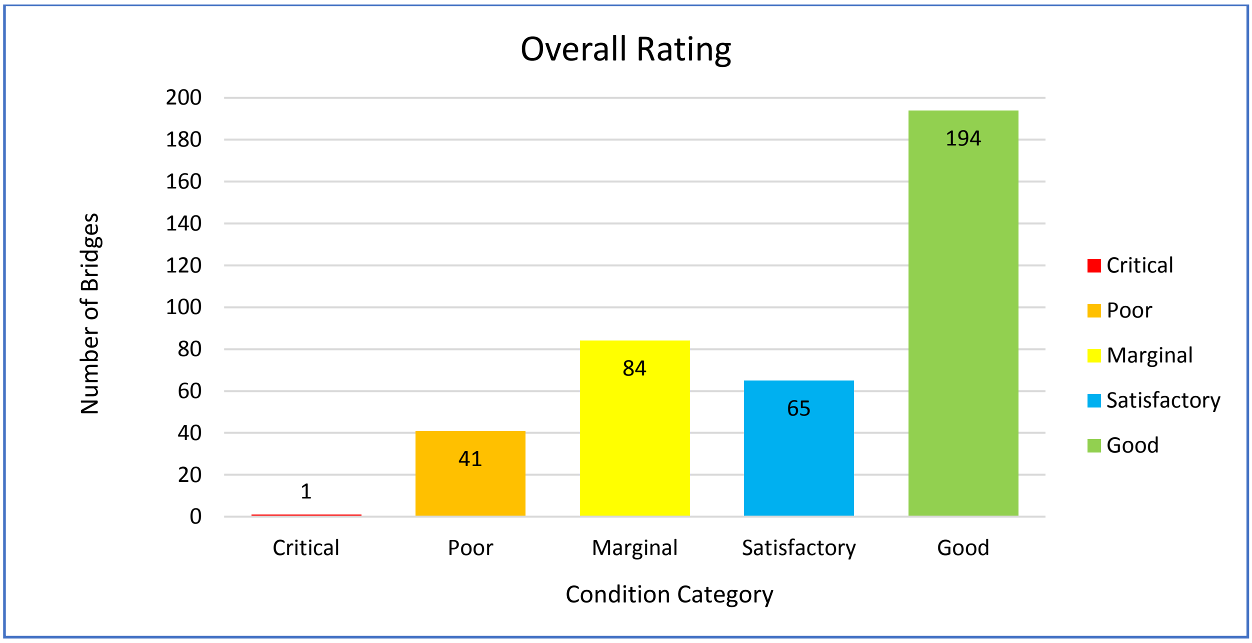


Figure 3-3: Overall rating of bridges

From Figure 3-3 above, it can be observed that the highest number of bridges (50%) is in good condition. The bridges in marginal condition are significant (17%) and timely maintenance is advised to prevent them from quickly deteriorating to a poor condition. Bridges in poor and critical condition are the least (11%) and urgent intervention is required.

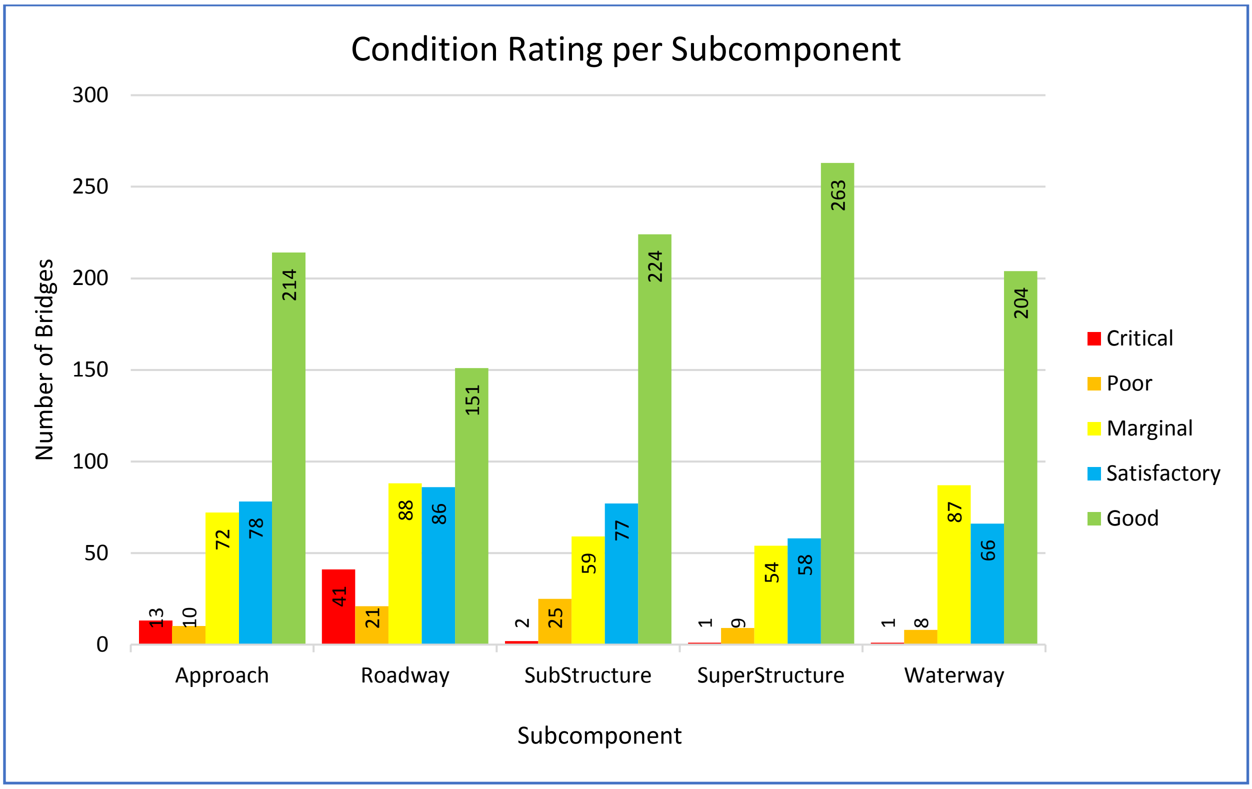


Figure 3-4: Condition rating of bridges per subcomponent

From the figure above, it is observed that;

- The majority of bridges have subcomponents in good condition.
- An appreciable number of subcomponents have a marginal rating and timely maintenance is required.
- Subcomponents in poor and critical condition are minimal and urgent maintenance is required. Failure to do so possess a safety hazard and can lead to closure or collapse of a bridge.

3.2 Condition of Major Culverts

Inventory and condition surveys were carried out as stated in 3.1 above. A total of 253 (86%) Major Culverts on the network have condition assessment data. Based on these visual assessments, the following conclusions can be made with regards to the condition of Major Culverts in UNRA.

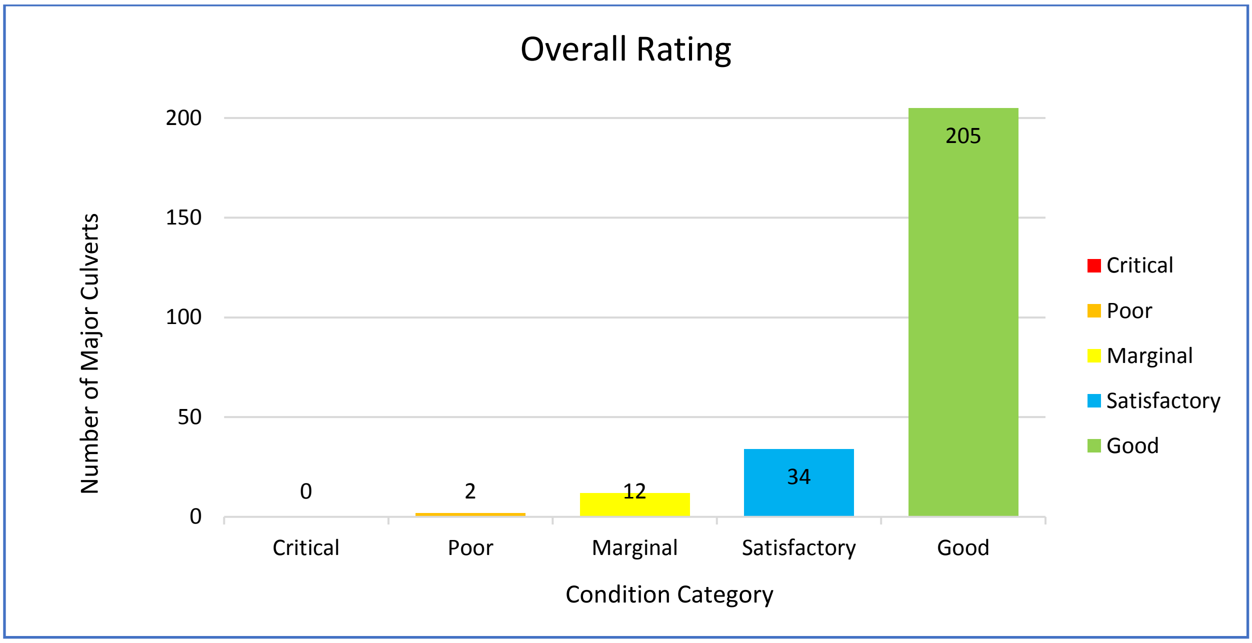


Figure 3-5: Overall rating of major culverts

From Figure 3-3 above, it can be observed that 205 (81%) major culverts are in good condition. 34 (13%) are satisfactory and 12 (5%) are in marginal condition. Only 2 (less than 1%) major culverts are in poor and critical condition.

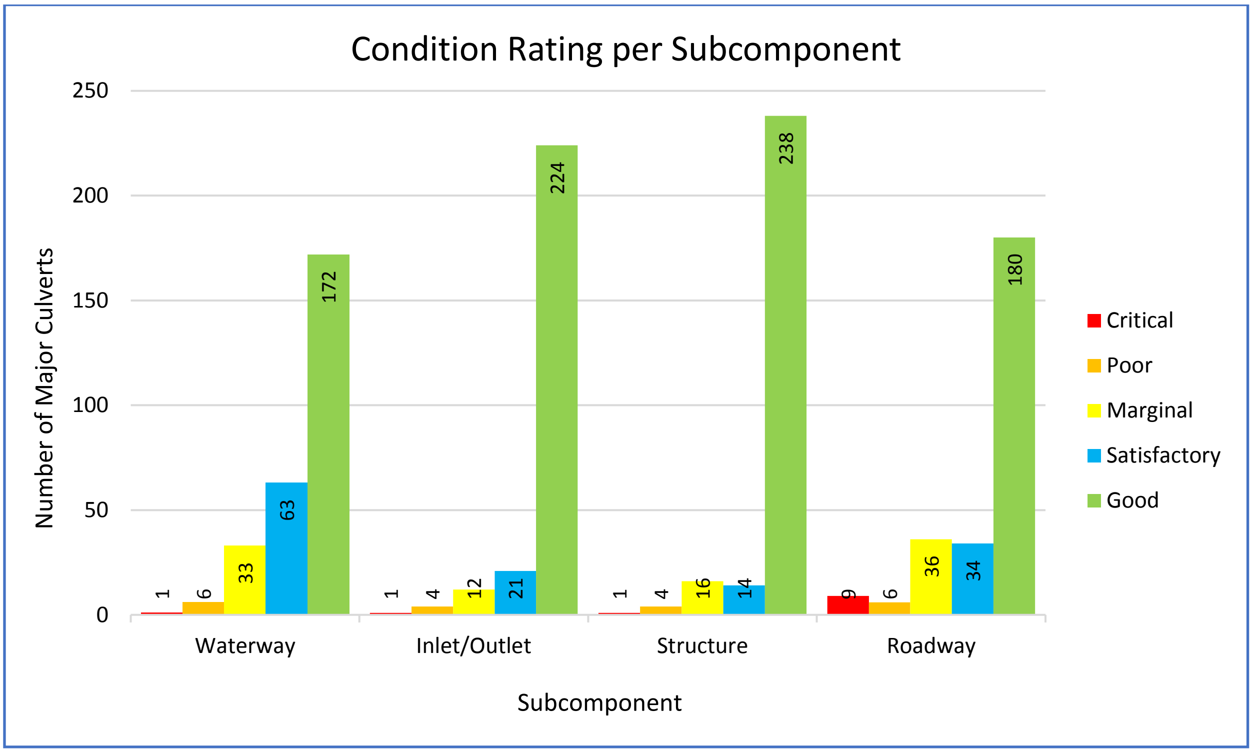


Figure 3-6: Condition rating of bridges per subcomponent

From the figure above, it is observed that;

- The majority of major culverts have subcomponents in good condition.

- Subcomponents in poor and critical condition are minimal and urgent maintenance is required.

4 CONCLUSION

The UNRA network has 727 structures with 434 bridges (60%) and 293 Major Culverts (40%). For these structures, there is visual condition assessment data available for 385 (88%) of Bridges and 253 (86%) of Major Culverts.

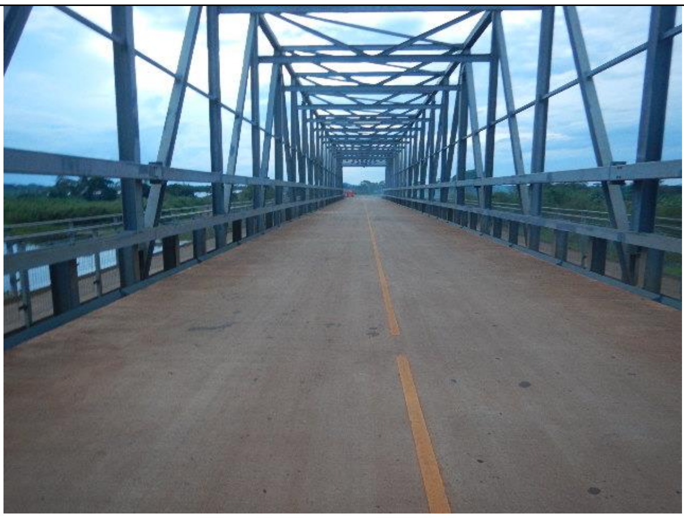
The condition of bridges is generally good with 50% in good condition, 17% satisfactory, 22% marginal, 11% poor and 0% in critical condition. Considering the condition of the subcomponents, the roadway has the highest number of bridges in critical condition 11%.

The condition of major culverts is generally good with 81% in good condition, 13% satisfactory, 5% marginal, 1% poor and 0% in critical condition. Considering the condition of the subcomponents, the roadway has the highest number of major culverts in critical condition 4%.

5 APPENDICES

5.1 Pictures of bridges on the network

 Aswa bridge along Atiak-Palabek in 2008	 Aswa bridge along Atiak-Palabek in 2018

 Kabaale Bridge along Kyankwazi – Ngoma in 2018	 Kabaale Bridge in 2018
 Culvert C171 along Kagamba – Ishaka in 2018	 Culvert C171 in 2008. Structure was formerly Kitagata bridge
 Culvert C193 along Kisinga-Kyarumba-Kibirizi in 2017	 Culvert C193 in 2008. Structure was formerly Dungulwa bridge

 Newly rehabilitated Puranga bridge along Ngetta - Puranga in 2018	 Old Puranga bridge in 2008 before rehabilitation
 Kochi bridge along Keriri – Midigo in 2018	 Poor condition of Kochi bridge
 C520 along Kitgum - Madi Opei	 Newly constructed box culvert C520

	
C404 along Bugambe-Kidoma is in poor condition	Underside of C404

5.2 List of Bridges on the Network

5.3 List of Major Culverts on the Network